

BLUESTOCK FINTECH PVT. LTD.

DATA ANALYST INTERNSHIP PROJECT REPORT

MUTUAL FUND ANALYTICS PLATFORM

End-to-End Data Engineering, Quantitative Risk Engine & BI Dashboard

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1. Executive Summary & Project Objectives

1.1 Business Context

The Indian mutual fund industry has experienced an unprecedented surge, managing over Rs. 81 lakh crore in Assets Under Management (AUM) across 1,908 distinct schemes and 26.12 crore investor folios as of December 2025. Driven by a deepening equity culture, monthly Systematic Investment Plan (SIP) inflows reached an all-time high of Rs. 31,002 crore. However, retail investors and financial advisors frequently struggle with fragmented public data formats and unquantified portfolio risk profiles.

This project delivers a centralized financial data platform that standardizes multi-source ingestion, calculates risk-adjusted performance metrics, and exposes interactive business intelligence layers to enable data-driven fund selections across India's growing investment landscape.

Metric	Value	Significance
Total Industry AUM	Rs. 81 Lakh Crore	Largest AUM in Indian financial history
Active Fund Schemes	1,908 Schemes	Covers all SEBI-regulated categories
Total Investor Folios	26.12 Crore	Reflects rapid retail investor penetration
Monthly SIP Inflows	Rs. 31,002 Crore	All-time high as of December 2025

Table 1.1 — Industry-Level Key Performance Indicators (December 2025)

1.2 Core Technical Objectives

The platform was engineered around five pillars of analytical capability, each addressing a specific gap in the retail and institutional investment analytics workflow:

- Automated Data Ingestion:** Build a robust Python ETL pipeline to ingest raw public AMFI, NSE, and open API datasets, ensuring schema validation, deduplication, and audit-trail logging at each transformation stage.
- Relational Persistence:** Design and populate a normalized relational SQLite database running a 5-table star schema optimized for financial analytical querying with sub-second cross-join response times.
- Quantitative Risk Core:** Programmatically calculate rolling 3-year CAGR, Sharpe ratios, Sortino ratios, information tracking errors, and market Alpha/Beta regressions using daily NAV history.
- Advanced Portfolio Diagnostics:** Implement parametric 95% Historical Value at Risk (VaR), Conditional VaR (CVaR), and Herfindahl-Hirschman Index (HHI) models for portfolio concentration audits.
- Behavioral Cohort Analytics:** Group transactional registries to map investor retention patterns and automatically flag inactive, at-risk accounts using configurable behavioral thresholds.

2. Multi-Source Ingestion & Database Architecture

2.1 Ingestion Vectors

The platform safely handles a dataset inventory exceeding 87,000 transaction and valuation rows across 10 structured assets. Each source file underwent schema validation, null-value imputation, and type coercion before being committed to the relational store. The six primary ingestion sources are:

Source File	Volume	Description
01_fund_master.csv	40 Schemes	Descriptive dimensions for 40 real SEBI-registered fund schemes
02_nav_history.csv	64,000+ Rows	Daily historical NAV entries spanning January 2022 to May 2026
03_aum_by_fund_house.csv	Quarterly	Quarterly asset footprints for India's 10 largest fund houses
04_monthly_sip_inflows.csv	Monthly	Industry-wide systemic inflow metrics tracked over 54 months
08_investor_transactions.csv	21,000+ Rows	Transaction logs (SIP, Lumpsum, Redemptions) for 5,000 retail investors
10_benchmark_indices.csv	Daily Series	Daily market closing matrices for Nifty 50 and Nifty 100 tracking series

Table 2.1 — Platform Data Source Inventory

2.2 Relational Database Schema (schema.sql)

To enable sub-second cross-joins across deep transaction tables, data was loaded into a standardized SQLite database via SQLAlchemy. The schema follows a star schema design with two dimension tables (dim_fund, dim_date) and two fact tables (fact_nav, fact_transactions), optimized for analytical query patterns common in financial reporting workloads.

```
-- DDL Schema Layout for SQLite Relation Core
CREATE TABLE dim_fund (
    amfi_code      TEXT      PRIMARY KEY,
    fund_house     TEXT      NOT NULL,
    scheme_name    TEXT      NOT NULL,
    category       TEXT      NOT NULL,
    sub_category   TEXT      NOT NULL,
    expense_ratio_pct REAL    NOT NULL,
    risk_category  TEXT      NOT NULL
);
CREATE TABLE fact_nav (
    amfi_code      TEXT      REFERENCES dim_fund(amfi_code),
    date_id        TEXT      REFERENCES dim_date(date_id),
    nav            REAL      NOT NULL,
```

```

        daily_return    REAL,
        PRIMARY KEY (amfi_code, date_id)
    );
CREATE TABLE fact_transactions (
    tx_id                TEXT    PRIMARY KEY,
    investor_id          TEXT    NOT NULL,
    amfi_code            TEXT    REFERENCES dim_fund(amfi_code),
    date_id              TEXT    REFERENCES dim_date(date_id),
    amount_inr           INTEGER NOT NULL,
    transaction_type     TEXT    NOT NULL, -- SIP | Lumpsum | Redemption
    city_tier            TEXT    NOT NULL, -- T30 | B30
    kyc_status           TEXT    NOT NULL
);

```

Listing 2.1 — Core DDL Schema Layout (schema.sql)

2.3 Data Dictionary

The following data dictionary documents all field-level specifications across the four primary schema objects, providing type definitions, constraints, and semantic descriptions for data governance purposes.

Table	Column	Type	Constraint	Description
dim_fund	amfi_code	TEXT	PK	Unique AMFI scheme identifier
dim_fund	fund_house	TEXT	NOT NULL	Asset Management Company name
dim_fund	scheme_name	TEXT	NOT NULL	Full SEBI-registered scheme name
dim_fund	category	TEXT	NOT NULL	SEBI category: Equity / Debt / Hybrid
dim_fund	expense_ratio_pct	REAL	NOT NULL	Annual total expense ratio (%)
dim_fund	risk_category	TEXT	NOT NULL	SEBI riskometer classification
dim_date	date_id	TEXT	PK	Surrogate date key (YYYY-MM-DD format)
dim_date	calendar_date	DATE	NOT NULL	Actual calendar date value
dim_date	year / month	INTEGER	NOT NULL	Calendar year and month integers
dim_date	quarter	TEXT	NOT NULL	Fiscal quarter label (e.g., Q1-FY26)
dim_date	is_weekday	INTEGER	0 or 1	Boolean flag: 1 = trading day
fact_nav	amfi_code	TEXT	FK / PK	Foreign key to dim_fund (composite PK)
fact_nav	date_id	TEXT	FK / PK	Foreign key to dim_date (composite PK)
fact_nav	nav	REAL	NOT NULL	Net Asset Value per unit (INR)
fact_nav	daily_return	REAL	Nullable	Computed: (NAV_t / NAV_t-1) - 1

Table	Column	Type	Constraint	Description
fact_transactions	tx_id	TEXT	PK	UUID transaction identifier
fact_transactions	investor_id	TEXT	NOT NULL	Anonymized investor registry ID
fact_transactions	amount_inr	INTEGER	NOT NULL	Transaction amount in Indian Rupees
fact_transactions	transaction_type	TEXT	NOT NULL	SIP / Lumpsum / Redemption
fact_transactions	city_tier	TEXT	NOT NULL	T30 (urban) or B30 (regional)
fact_transactions	kyc_status	TEXT	NOT NULL	KYC compliance flag (Verified / Pending)

Table 2.2 — Complete Data Dictionary: Schema Objects and Field Specifications

3. Exploratory Data Analysis & Statistical Highlights

3.1 Macro Inflow Trajectories

Exploratory analysis confirms the structural dominance of top Asset Management Companies. SBI Mutual Fund maintains market leadership with an active asset base of Rs. 12.50 lakh crore, closely shadowed by ICICI Prudential MF at Rs. 10.74 lakh crore and HDFC MF at Rs. 9.30 lakh crore.

Time-series analyses of inflows reveal a massive structural shift toward systemic equity culture. Retail systematic investment patterns peaked alongside broader market indices, pushing total industry folios to 26.12 crore entries and validating the urgency of analytics tooling at scale.

#	Asset Management Company	AUM (Rs. L. Crore)	YoY	Segment
1	SBI Mutual Fund	12.50	+14.2%	Market Leader
2	ICICI Prudential Mutual Fund	10.74	+11.8%	2nd Largest AMC
3	HDFC Mutual Fund	9.30	+9.5%	3rd Largest AMC
4	Nippon India Mutual Fund	7.20	+16.1%	Fast Growing
5	Kotak Mahindra Mutual Fund	6.85	+8.3%	Consistent Growth

Table 3.1 — Top 5 AMCs by Assets Under Management (December 2025)

3.2 Demographic and Geographic Segmentations

Geospatial profiles indicate that while Top 30 (T30) urban capital centers provide approximately 60.15% of immediate transactional volumes, Beyond 30 (B30) regional territories represent the fastest growing retail onboarding segment. The platform captures this bifurcation through the city_tier field in fact_transactions, enabling granular policy segmentation.

Age cohort groupings confirm that digital-native retail investors within the 18-35 age bracket make up over 42% of modern monthly transactional activity, emphasizing the need for streamlined, digital-first analytics interfaces and mobile-optimized reporting layers.

Segment	Category	Value	Platform Insight
Geographic	T30 Urban	60.15%	Volume-dominant; near-saturation growth
Geographic	B30 Regional	39.85%	Fastest-growing retail onboarding pipeline
Age Cohort	18-35 Years	42%+	Digital-first; SIP-dominant behavior
Age Cohort	36-55 Years	~38%	Highest per-ticket lumpsum values
Transaction Type	SIP Contributions	~55%	Dominant systematic inflow mechanism
Transaction Type	Lumpsum	~30%	Bulk institutional & HNI inflows

Table 3.2 — Demographic & Geographic Segmentation Summary

4. Quantitative Analytics & Performance Modeling

4.1 Risk-Adjusted Calculations

Using daily historical valuation paths, the performance engine computed annualized multi-period Compound Annual Growth Rates (CAGR). To isolate true management skill from broad market movements, individual asset tracks were regressed against the Nifty 100 benchmark using Ordinary Least Squares (OLS) estimation. The following formulas form the analytical backbone of the risk engine:

Core Risk-Adjusted Return Formulas

(1) Daily Return: $\text{daily_return}(t) = [\text{NAV}(t) / \text{NAV}(t-1)] - 1$

(2) CAGR (3-Year): $\text{CAGR} = [(\text{NAV_end} / \text{NAV_start})^{(1/3)}] - 1$

(3) Sharpe Ratio: $\text{Sharpe} = (R_p - R_f) / \sigma_p$

(4) Sortino Ratio: $\text{Sortino} = (R_p - R_f) / \sigma_{\text{downside}}$

(5) Alpha (OLS): $\alpha = R_{\text{fund}} - [R_f + \beta \times (R_{\text{market}} - R_f)]$

Where: R_p = portfolio return | R_f = 6.50% (RBI repo rate proxy) | σ_{downside} = std. dev. of negative daily returns only

Formulas 4.1 — Core Risk-Adjusted Return Metrics

The risk-free proxy (R_f) was anchored securely at 6.50%, representing the prevailing RBI repo rate standard. Downside risk deviation (σ_{downside}) exclusively considers trading entries with net-negative daily returns,

ensuring accurate asymmetric downside risk measurements that penalize volatility only in adverse market conditions.

4.2 Composite Scorecard Evaluation

A multi-factor scoring model compiled asset metrics into a 0-100 quality matrix using weighted percentiles. The composite scoring engine normalizes each metric to a 0-100 scale using the fund universe as a peer group, then applies the following weighting structure to arrive at a final quality score:

Composite Fund Quality Score (0-100)

$$\text{Composite Score} = 0.30 \times R(3Yr) + 0.25 \times \text{Sharpe} + 0.20 \times \text{Alpha} + 0.15 \times \text{Expense_Inv} + 0.10 \times \text{Drawdown_Inv}$$

Expense_Inv and Drawdown_Inv are inverted metrics (lower is better), normalized to 0-100 within the peer universe

Formula 4.2 — Multi-Factor Composite Scoring Model

Factor	Metric	Weight	Rationale
Performance	3-Year Rolling CAGR	30%	Primary return driver
Risk-Adjusted Return	Sharpe Ratio	25%	Risk-normalized quality
Manager Skill	Market Alpha (OLS)	20%	Excess return attribution
Cost Efficiency	Expense Ratio (Inv)	15%	Lower cost = higher score
Downside Protection	Max Drawdown (Inv)	10%	Capital preservation quality

Table 4.1 — Composite Scorecard Factor Weights and Rationale

Top alpha-generating portfolios extracted by the scoring engine show high performance resilience. Aggressive equity models consistently outpaced benchmark indices (Nifty 50 and Nifty 100), producing notable positive alphas while keeping market sensitivities (Betas) within an acceptable range of 0.85 to 1.22.

Fund Name	3Y CAGR	Sharpe	Alpha	Beta	Score
Mirae Asset Large Cap Fund	18.4%	1.42	+2.31%	0.94	87.2
SBI Small Cap Fund	24.7%	1.18	+3.85%	1.08	84.6
Axis Bluechip Fund	16.2%	1.35	+1.92%	0.91	82.1
Parag Parikh Flexi Cap	19.8%	1.29	+2.67%	0.88	79.8
HDFC Mid-Cap Opportunities	22.1%	1.11	+1.74%	1.12	77.4

Table 4.2 — Top 5 Funds by Composite Scorecard (fund_scorecard.csv snapshot)

5. Advanced Risk Modeling & Behavioral Analytics

5.1 Tail Risk Measurement: Parametric VaR & CVaR

To model portfolio risks during deep market corrections, a tail-risk module computes 95% Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) across every active schema. By sampling the 5th percentile of historical daily log-return sets, the system quantifies worst-case drawdown boundaries that can be used to inform portfolio stress-testing and capital allocation decisions.

Tail Risk Formulas: VaR and CVaR

(6) $\text{VaR}(95\%)$: $\text{VaR}_{95\%} = \text{Percentile}(\Delta\text{NAV}_{\text{daily}}, 5)$

(7) $\text{CVaR}(95\%)$: $\text{CVaR}_{95\%} = E[\Delta\text{NAV}_{\text{daily}} | \Delta\text{NAV}_{\text{daily}} \leq \text{VaR}_{95\%}]$

CVaR represents the expected loss given that the loss exceeds the VaR threshold (Expected Shortfall)

Formulas 5.1 — Parametric VaR and Conditional Value at Risk

Fund Category	Daily VaR (95%)	CVaR (95%)	Volatility (σ)	Risk Classification
Small Cap Equity	-2.85%	-3.42%	1.92%	Very High Risk
Mid Cap Equity	-2.11%	-2.68%	1.54%	High Risk
Large Cap Equity	-1.58%	-1.94%	1.12%	Moderate Risk
Balanced / Hybrid	-1.02%	-1.28%	0.78%	Low-Moderate Risk
Liquid / Debt Fund	-0.08%	-0.11%	0.04%	Very Low Risk

Table 5.1 — VaR/CVaR Summary by Fund Category (var_cvar_report.csv)

5.2 Portfolio Concentration Audits (HHI Index)

The system calculates the Herfindahl-Hirschman Index (HHI) across sectoral allocations to monitor concentration risk within individual fund portfolios. The HHI is a standard economic measure of market concentration, adapted here for mutual fund sector exposure monitoring:

Herfindahl-Hirschman Index (Portfolio Concentration)

(8) $\text{HHI} = \sum_i (W_{\text{sector},i})^2$ where $i = 1$ to N sectors

HHI > 0.25 triggers concentration warning flag (heavy BFSI/Tech exposure); HHI < 0.10 = well-diversified

Formula 5.2 — Herfindahl-Hirschman Index for Sector Concentration

Portfolios displaying an HHI index greater than 0.25 indicate high sectoral exposure (typically heavily weighted in BFSI and Technology blocks), prompting systematic warning flags within the recommendation pipeline. The engine currently monitors 40 fund portfolios in near-real-time against this threshold.

5.3 Retention Analysis & Churn Modeling

Analyzing transaction intervals via `sip_continuity.csv` revealed key behavioral patterns that inform the platform's investor retention module. The system automatically classifies investors into three behavioral cohorts based on their SIP contribution history and transaction gap analysis:

Cohort Label	Classification Criteria	Risk Level	Platform Action
Active Investor	Gap < 35 calendar days	Low	Standard engagement; monthly reporting
At-Risk Account	Gap between 35-90 days	Medium	Automated re-engagement trigger deployed
Lapsed Investor	Gap > 90 days	High	Distributor alert + incentive pathway

Table 5.2 — Investor Behavioral Cohort Classification Framework

Investors with systemic account gaps exceeding 35 calendar days are automatically flagged as "At-Risk Accounts". This allows distributors and platform market engineers to deploy targeted automated re-engagement tools before full capital redemptions occur, directly reducing churn-related AUM erosion.

6. Business Intelligence Architecture

The front-end interactive system contains a multi-page dashboard built using Power BI Desktop, connecting directly to the normalized SQL storage framework via DirectQuery and scheduled ODBC refresh modes. The dashboard was designed to serve multiple user personas: portfolio managers, compliance officers, and retail advisor channels.

Page	Dashboard Name	Key Visuals	Business Purpose
1	Industry Overview	KPI cards, AMC ranking bar charts, AUM trend lines	Executive-level snapshot: Total AUM (Rs. 81L Cr), monthly inflow bounds, multi-year AMC trajectory rankings
2	Fund Performance Analytics	Risk vs. Return scatter plots, benchmark overlays, drilldown filters	Interactive fund comparison by category and fund house; identifies alpha-generating vs. index-hugging funds

Page	Dashboard Name	Key Visuals	Business Purpose
3	Investor Demographics	Choropleth maps, age pyramid charts, city-tier stacked bars	Visualizes transactional volumes across city tiers, age distributions, and geographic maps for market segmentation
4	Market Trend Matrix	Dual-axis SIP vs. benchmark line chart, correlation heatmap	Connects systematic retail inflow registries against Nifty 50 / Nifty 100 baseline trajectories for macro analysis

Table 6.1 — Power BI Dashboard Page Architecture

Data Connectivity Architecture

- SQLite database exposed via ODBC driver layer for Power BI DirectQuery mode
- Scheduled data refresh triggers nightly ETL pipeline re-runs at 00:30 IST
- Row-level security (RLS) profiles isolate AMC-specific views for fund house partners
- Custom DAX measures replicate Python-side Sharpe, Alpha, and VaR calculations for cross-validation

Note 6.1 — Power BI Data Connectivity and Security Architecture

7. Operational Blueprint & Platform Deployment

7.1 Automated Orchestration (run_pipeline.py)

System updates are fully automated using a master pipeline controller (run_pipeline.py). When triggered on a schedule or via manual invocation, the system runs end-to-end extraction, cleaning, analytical updates, and automated parameter matching sequentially. Each module logs its execution status to a timestamped audit file for traceability.

```
[ STARTING BLUESTOCK MUTUAL FUND ANALYTICS MASTER PIPELINE ]
=====
Executing Script: scripts/live_nav_fetch.py...
  Contacting API For: HDFC Top 100 Direct (125497)... Saved Successfully.
  Contacting API For: SBI Bluechip (119551)... Saved Successfully.
Executing Notebook: notebooks/Advanced_Analytics.ipynb... Updated Cells.
Executing Script: scripts/recommender.py... Querying optimal allocations.
=====
[ COMPLETED AUTOMATION HUB RUN EXECUTION - RELEASE V1.0 ]
```

Listing 7.1 — Automated Pipeline Execution Log (run_pipeline.py console output)

Step	Module	Output Artifact	Description
1	live_nav_fetch.py	nav_history_updated.csv	Fetches latest NAV from AMFI public API

Step	Module	Output Artifact	Description
2	etl_pipeline.py	analytics.sqlite	Ingests, validates, and loads all 6 sources
3	Advanced_Analytics.ipynb	fund_scorecard.csv, var_cvar_report.csv	Computes CAGR, Sharpe, Alpha, VaR, CVaR, HHI
4	recommender.py	recommendations.json	Ranks funds by composite score for user profiles
5	Power BI Refresh	Dashboard (4 pages)	Auto-refreshes BI layer from updated SQLite

Table 7.1 — Automated Pipeline Stage Registry

7.2 Recommendations for Future Development

The current platform represents a robust analytical foundation. The following roadmap items have been identified to scale the system to production-grade infrastructure and enhanced analytical capability:

Initiative	Priority	Technical Description
Production Scaling	High	Migrate the development-grade SQLite layer to a distributed cloud PostgreSQL framework (AWS RDS or GCP CloudSQL) to support concurrent write threads, connection pooling, and enterprise-grade backups
Real-Time Data Streams	High	Integrate Apache Kafka streaming architectures to process immediate public intraday transaction adjustments rather than relying on daily file-batch ingestion cycles
Predictive Risk Engine	Medium	Implement LSTM and GRU recurrent neural network models to forecast NAV trajectories and supplement current parametric historical risk frameworks with forward-looking probabilistic estimates
REST API Layer	Medium	Expose a FastAPI microservice layer over the risk engine, enabling third-party fintech apps and advisor portals to query fund scores, VaR metrics, and recommendations programmatically
NLP Regulatory Alerts	Low	Build a natural language processing module to parse SEBI circulars and AMC scheme information documents (SIDs) for automated compliance flag generation

Table 7.2 — Platform Development Roadmap

This project demonstrates end-to-end mastery of data engineering, quantitative finance modeling, and business intelligence design within the Indian mutual fund ecosystem. The platform architecture is designed for extensibility, and the modular codebase enables seamless integration of future enhancements without disrupting existing analytical pipelines. All deliverables, from the normalized relational database to the

interactive Power BI dashboards, have been validated against real AMFI public data sources and are ready for stakeholder demonstration.

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